

Conditional Davenport Corridors and Saturated Obstructions in C_5^3

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25 August 2026

Abstract

Let $D_k(G)$ be the least length that forces k pairwise disjoint nonempty zero-sum subsequences. Freeze and Schmid proved the recurrence and the lower line $D_k(C_5^3) \geq 5k + 10$; Fan et al. record $s_{\leq 5}(C_5^3) = 33$ and a support-eight inverse property attributed to Gao et al. We derive $D_2(C_5^3) = 20$ from Freeze–Schmid’s lower bound and Zhao’s short-subsequence lemma. Assuming only $D_3(C_5^3) = 25$ and $s_{\leq 6}(C_5^3) = 24$, we obtain $5k + 10 \leq D_k(C_5^3) \leq 5k + 11$ for every $k \geq 4$.

If $D_4(C_5^3) = 30$, then the lower line is exact for every $k \geq 2$. To analyze the remaining alternative, we study a hypothetical total-zero sequence of length 31 with no nonempty zero-sum subsequence of length at most five. For saturated short-free sequences in odd-prime elementary abelian groups, we prove a defect certificate excluding multiplicity $p - 2$. For $p = 5$, a saturated obstruction has only multiplicities 1, 2, 4. If s is support size and c_i counts multiplicity i , then $c_2 = 31 - s - 3c_4$ and $c_1 = 2s - 31 + 2c_4$.

The subsequence of points of multiplicity at least two has length $62 - 2s - 2c_4$. Using the short-zero-sum threshold $\eta(C_5^2) = 13$, we prove that this stratum spans rank three whenever $s + c_4 \leq 24$. We also prove an exact identity coupling support deficit, repetitions inside zero-sum atoms, and overlaps between atom supports. Two exact implementations found no obstruction through support ten within the supplied computational model. Because this computation has not been independently replicated, we report it only as bounded computational evidence.

Under H1 and H2, these results leave $D_4(C_5^3) \in \{30, 31\}$.

Keywords: generalized Davenport constants; zero-sum sequences; elementary abelian groups; short zero sums; inverse zero-sum problems

1. Introduction

Generalized Davenport constants measure the sequence length required to force several pairwise disjoint zero sums. Rank-two groups exhibit strong eventual structure, whereas rank-three groups can depart from the expected arithmetic progression.

Under the declared inputs below, the values $D_2 = 20$ and $D_3 = 25$, together with short-zero-sum thresholds, reduce the next value to

$$D_4(C_5^3) \in \{30, 31\}.$$

This one-bit ambiguity controls the tail: the lower candidate $D_4 = 30$ forces $D_k = 5k + 10$ for every $k \geq 2$. The upper candidate would yield a rigid length-31 total-zero sequence with no zero sum of length at most five.

We combine the recurrence with symbolic analysis of that hypothetical obstruction. The purpose is twofold: to obtain a conditional width-one theorem for the full tail and to identify exactly where existing rank reductions cease to be automatic. Computation is kept in a separate evidence class.

Our contributions are:

1. the conditional width-one corridor for every $k \geq 4$;
2. the conditional exact tail if $D_4 = 30$;
3. an odd-prime saturation-defect lemma excluding multiplicity $p - 2$;
4. bounded computational evidence through support ten;
5. an exact multiplicity parameterization for every remaining support stratum;
6. the rank-forcing phase $s + c_4 \leq 24$ for the high-multiplicity stratum; and
7. an identity coupling support deficit to internal atom repetition and cross-atom overlap.

2. Definitions and declared inputs

For a finite abelian group G , $D_k(G)$ is the least integer n such that every sequence of length at least n over G contains k pairwise disjoint nonempty zero-sum subsequences. Let $s_{\leq \ell}(G)$ denote the least length forcing a nonempty zero-sum subsequence of length at most ℓ . For an odd prime p , a sequence is *p-short-free* if it has no nonempty zero-sum subsequence of length at most p . It is *saturated* relative to this property if appending any group element destroys it. Thus *5-short-free* and *saturated* below are the specializations at $p = 5$.

We use the generalized-Davenport recurrence of Freeze and Schmid [1, Proposition 3.1]

$$D_{k+1}(G) \leq \max\{D_k(G) + \ell, s_{\leq \ell}(G) - 1\},$$

and their lower bound $D_k(C_5^3) \geq 5k + 10$ [1, Theorem 4.1]. Fan et al. record $s_{\leq 5}(C_5^3) = \eta(C_5^3) = 33$ and Property C for C_5^3 [2, Lemma 4.2(5),(8)], attributing the original results to Gao et al. [6]. By the definition of Property C, every length-32 5-short-free sequence has exactly eight distinct support points. We also use the rank-two identity $\eta(C_m^2) = 3m - 2$, hence $\eta(C_5^2) = 13$ [2, p. 1].

For clarity, the two assumptions not established in the cited literature are

$$(H1) \quad D_3(C_5^3) = 25, \quad (H2) \quad s_{\leq 6}(C_5^3) = 24.$$

Every corridor or obstruction statement below that uses H1 or H2 says so explicitly. Neither assumption is presented as a theorem of this paper.

Proposition 1. $D_2(C_5^3) = 20$.

Proof. Freeze and Schmid's lower bound gives $D_2(C_5^3) \geq 20$. For the reverse inequality, Olson's formula for finite abelian p -groups gives $D(C_5^3) = 1 + 3(5 - 1) = 13$ [7]. Let S be an arbitrary sequence of length 20 and append $g = -\sigma(S)$, producing the zero-sum sequence $T = Sg$ of length 21. Apply Zhao's Lemma 4.4 [3] with $p = 5$, $D(C_5^3) = 13$, $|T| = 21$, $k = 8$, and $i = 2$. Its coefficient is $\binom{13}{6} + \binom{14}{7} = 5148 \equiv 3 \pmod{5}$, so T has a nonempty zero-sum subsequence of length at most seven. Its zero-sum complement has length at least 14 and therefore contains a nonempty proper zero-sum subsequence because $D(C_5^3) = 13$; the remaining complement is zero-sum as well. Thus T contains three pairwise disjoint nonempty zero-sum subsequences. At most one contains the

appended term g , so the other two are disjoint zero-sum subsequences of the arbitrary sequence S . Therefore $D_2(C_5^3) \leq 20$. $\square$

3. A width-one tail

Taking $\ell = 6$ gives

$$D_4 \leq \max\{25 + 6, 24 - 1\} = 31.$$

Together with the lower bound, this proves $D_4 \in \{30, 31\}$.

Theorem 2 (conditional corridor). Under H1 and H2, for every $k \geq 4$,

$$5k + 10 \leq D_k(C_5^3) \leq 5k + 11.$$

Proof. The base case is $D_4 \leq 31$. Apply the recurrence with $\ell = 5$ and $s_{\leq 5} - 1 = 32$. If $D_k \leq 5k + 11$, then both terms in the maximum are at most $5(k + 1) + 11$. Freeze and Schmid's lower bound supplies the lower line. $\square$

Theorem 3 (conditional tail). Under H1 and H2, if $D_4(C_5^3) = 30$, then

$$D_k(C_5^3) = 5k + 10 \quad (k \geq 2).$$

Proof. The statement holds for $k = 2, 3, 4$. Applying the same recurrence with $\ell = 5$ propagates the upper bound $5(k + 1) + 10$, which matches the published lower bound. $\square$

There is no converse here: the assumption $D_4 = 31$ does not by itself prove that later terms lie on the upper line.

4. The hypothetical length-31 obstruction

Assume H1 and H2, and suppose $D_4(C_5^3) = 31$. The extremal characterization of Freeze and Schmid provides a total-zero sequence S of length 31 whose maximum factorization length is four. If M is the minimum length of an atom dividing S , their recurrence gives $31 = D_4 \leq D_3 + M = 25 + M$, so $M \geq 6$. On the other hand, $s_{\leq 6}(C_5^3) = 24$ forces every 31-term sequence to contain a zero-sum subsequence of length at most six; an inclusion-minimal such subsequence is an atom. Hence $M = 6$, and S contains no zero-sum subsequence of length at most five. Thus the upper candidate yields a sequence satisfying

$$|S| = 31, \quad \sigma(S) = 0,$$

with no nonempty zero-sum subsequence of length at most five. Every element has multiplicity at most four. If its support exceeds eight, it is saturated: otherwise a short-free one-term extension would have length 32, whereas Property C [2] forces every length-32 5-short-free sequence to have exactly eight support points.

Theorem 4 (saturation defect). Let p be odd and let S be a saturated p -short-free sequence over an elementary abelian exponent- p group. If a nonzero element x has multiplicity $m < p$, then there is a subsequence R such that

$$|R| \leq p - 1 - m, \quad x \notin \text{supp}(R), \quad \sigma(R) = -(m + 1)x.$$

Proof. Append one further copy of x . Saturation supplies a zero-sum subsequence of length at most p that uses the appended occurrence. It must also use every original copy of x ; otherwise replacing the appended copy by an unused original occurrence would give a short zero sum already contained in S . Removing the $m + 1$ copies of x leaves the required R . $\square$

Corollary 5. Multiplicity $p - 2$ is impossible.

Indeed, the defect subsequence would have length at most one and sum x , forcing a forbidden additional copy of x . For $p = 5$, multiplicity three is absent. Singleton and double points also acquire defect certificates of length at most three and two, respectively.

5. Exact multiplicity grammar

For the saturated case $s \geq 9$, Corollary 5 excludes multiplicity three. Let $s = |\text{supp}(S)|$, and let c_1, c_2, c_4 count points of multiplicity one, two, and four. Then

$$c_1 + c_2 + c_4 = s, \quad c_1 + 2c_2 + 4c_4 = 31.$$

Writing $c_4 = j$ gives

$$c_2 = 31 - s - 3j, \quad c_1 = 2s - 31 + 2j.$$

Thus c_1 is odd, and nonnegativity determines the complete admissible range for every support size. For example, the support-23 patterns are

$$1^{15}2^8, \quad 1^{17}2^54, \quad 1^{19}2^24^2,$$

and the support-24 patterns are

$$1^{17}2^7, \quad 1^{19}2^44, \quad 1^{21}24^2.$$

6. The rank-forcing phase

Let H be the subsequence consisting of all points of multiplicity two or four. Its length is

$$|H| = 2c_2 + 4c_4 = 62 - 2s - 2c_4.$$

If H were contained in a subgroup of rank at most two, it would be a 5-short-free sequence over C_5^2 . The value $\eta(C_5^2) = 13$ [2, p. 1] forces a short zero sum once $|H| \geq 13$. Since $|H|$ is even, the contradiction begins at $|H| \geq 14$.

Theorem 6 (rank-forcing phase). For a saturated obstruction with $s \geq 9$, the high-multiplicity stratum spans rank three whenever

$$s + c_4 \leq 24.$$

Proof. The condition $|H| \geq 14$ is equivalent to $62 - 2s - 2c_4 \geq 14$, hence to $s + c_4 \leq 24$. $\square$

The theorem gives the following exact boundary:

Support	c_4	Consequence from Theorem 6
23	0 or 1	rank three forced
23	2	threshold is silent
24	0	rank three forced
24	1 or 2	threshold is silent
at least 25	any admissible value	threshold is silent

Silence is not a converse. Outside the phase, rank three may follow from a different argument.

7. Bounded low-support exclusion

Symbolic reductions combined with two exact state representations exclude every support size through ten.

Computational result. In the supplied exact search model, no length-31 total-zero 5-short-free sequence over C_5^3 exists with support at most ten.

Exact search description. Multiplicity at most four first excludes supports at most seven. At support eight, the only length pattern is $4^7 3$. Adding the missing fourth copy of the triple point preserves short-freeness; total sum then forces that point to equal the negative support sum. After normalizing a basis by $\text{GL}(3, 5)$, two exact subset-sum representations enumerate all 564 normalized supports and find none satisfying this condition.

For supports nine and ten, saturation and Corollary 5 exclude multiplicity three. The support-nine equations leave only $4^7 21$; a canonical search with the last point forced by total sum visits 6,537,270 states in each of two distinct exact-weight subset-sum engines and finds no solution. At support ten, the only patterns are $1^3 4^7$ and $1, 2^3 4^6$. Four multiplicity-four points must span rank three because 16 terms in a rank-two subgroup would contradict $\eta(C_5^2) = 13$ [2, p. 1]. Normalizing an independent triple to the standard basis makes the remaining enumeration complete. The two engines agree exactly: they visit respectively 210,700 states with 3,558 leaves and 272,119 states with no leaves, finding zero solutions in both patterns.

Both engines prune a partial candidate exactly when adding a term creates a zero-sum subsequence of one of the lengths one through five; one stores explicit byte reachability by weight and group sum, while the other stores translation masks in a 128-bit representation. Their source, build instructions, expected rows, and machine-readable results are included as ancillary files. Thus every support stratum through ten is exhausted within the declared model. Because a clean-room external replay has not been completed, this result is not used in any analytic proof.

8. Atom repetition and overlap

Let $S = U_1 \cdots U_r$ be an occurrence-disjoint atom factorization. For each group element g , let r_g be the number of atom supports containing g , and define the internal deficit

$$\delta_i = |U_i| - |\text{supp}(U_i)|.$$

Theorem 7 (repetition-overlap identity). Every occurrence-disjoint factorization satisfies

$$|S| - |\text{supp}(S)| = \sum_i \delta_i + \sum_{g \in \text{supp}(S)} (r_g - 1).$$

Proof. Summing the atom support sizes counts every global support point r_g times. Hence

$$\sum_i \delta_i = |S| - \sum_g r_g,$$

while cross-atom overlap equals

$$\sum_g r_g - |\text{supp}(S)|.$$

Adding the two identities proves the claim. $\square$

Thus any independently established lower support bound s_0 gives a total internal-repetition plus cross-overlap budget of at most $31 - s_0$. The two packaged support-ten implementations suggest $s_0 = 11$ and hence budget 20, but that numerical consequence remains bounded computational evidence and is not used in any proof.

9. The residual obstruction phase

A decisive next analysis should impose simultaneously:

1. the multiplicity pattern from Section 5;
2. the rank status from Theorem 6;
3. the saturation-defect certificate at every singleton and double point;
4. the repetition-overlap identity; and
5. total sum zero with exclusion of zero sums of lengths one through five.

This intersection is substantially smaller than an unstructured support search. A surviving sequence would require independent factorization verification before establishing $D_4 = 31$. A complete, independently replayed infeasibility proof for the length-31 target would establish the lower candidate $D_4 = 30$.

10. Relation to prior work

The generalized-Davenport recurrence, lower-bound framework, extremal factorization language, and eventual-linearity context are established results [1]. The value $D_2(C_5^3) = 20$ is a direct corollary of the published lower bound and Zhao’s short-subsequence lemma [3], not a novelty claim here. The C_0 framework and short-zero-sum localization are also prior work [2], as are the inverse zero-sum program [4] and rank-two inverse results [5].

The residual contribution is the width-one synthesis for C_5^3 , the saturation-defect specialization, the exact multiplicity grammar, the rank-forcing phase, and the coupling of support deficit with atom repetition and overlap. These are structural compression results; they do not decide the final extremal bit.

11. Reproducibility and limitations

The conditional corridor, defect lemma, and support-ten exclusion are separately bound to exact records. A separate implementation enumerates all multiplicity patterns for supports 8-31, checks the rank-phase equivalence, reproduces the support-23 and support-24 tables, and validates the atom identity on independently generated factorizations. Only the displayed analytic proofs carry all-parameter authority.

The exact value of $D_4(C_5^3)$ remains open, as does the associated length-31 extremal-spectrum statement. The rank phase is one-way, and the atom identity is a compression principle rather than a contradiction. No additional implementation can substitute for an independently auditable final obstruction proof.

12. Conclusion

Conditional on H1 and H2, the generalized Davenport constants of C_5^3 remain within one unit of the line $5k + 10$ for every $k \geq 4$, and the lower choice at $k = 4$ would fix the entire tail. The hypothetical upper-line obstruction has substantially more structure than this corridor alone reveals: its multiplicities have an exact grammar, its high-multiplicity stratum enters a sharp rank-forcing phase, and its atom factorization obeys a global repetition-overlap budget.

These results explain where the residual difficulty begins and define a smaller intersection problem for the unresolved extremal bit. They do not claim to have resolved it.

Tool-use disclosure

A generative language model assisted manuscript organization, language revision, and submission-package preparation. The listed author remains responsible for the mathematical statements, proofs, references, executable claims, and final text.

Data and code availability

The verification package contains standalone sources, build instructions, and expected results for the bounded support-eight-to-ten replay. The analytic multiplicity, rank-phase, and atom-identity arguments are contained in the manuscript; no separate executable verification is claimed for them.

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